

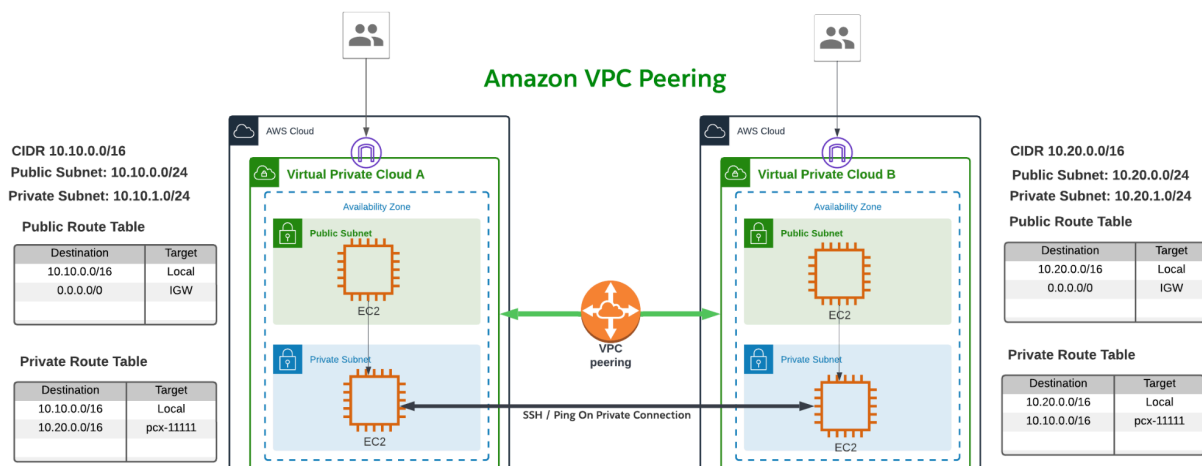
aws vpc inter-region peering

What is VPC Peering?

VPC Peering is a networking connection between two Virtual Private Clouds (VPCs) that allows them to **communicate privately** using private IP addresses over the AWS internal network.

It creates a **secure private connection** between two VPCs without using the internet.

VPC Peering enables secure, low-latency private communication between two VPCs using AWS's internal network.



Common Use Cases

- Connecting application VPC to database VPC
- Separating dev / test / prod environments
- Sharing common services like monitoring or logging

Important Limitations (Good to Mention)

- CIDR blocks must not overlap
- VPC peering is not transitive
- Manual route table configuration is required

Benefits of VPC Peering

- 🔒 1. High Security
- ⚡ 2. Low Latency & High Performance
- 💰 3. Cost-Effective
- 🧩 4. Simple Architecture
- 🌐 5. Cross-Account & Cross-Region Support
- 🔄 6. Uses Private IP Communication

steps...

Create a vpc

Vpc-1

cidr(classless inter domain name)=10.0.0.0/16 (65,536) ip address

Tendency=default

Create...

The screenshot shows the 'VPC settings' section of the AWS VPC console. It includes the following fields and options:

- Resources to create:** Two radio buttons. 'VPC only' is selected. 'VPC and more' is unselected.
- Name tag - optional:** A text input field containing 'vpc-1'. Below it is a small note: 'Creates a tag with a key of 'Name' and a value that you specify.'
- IPv4 CIDR block:** Two radio buttons. 'IPv4 CIDR manual input' is selected. 'IPAM-allocated IPv4 CIDR block' is unselected.
- IPv4 CIDR:** A text input field containing '10.0.0.0/16'. Below it is a small note: 'CIDR block size must be between /16 and /28.'
- IPv6 CIDR block:** Four radio buttons. 'No IPv6 CIDR block' is selected. 'IPAM-allocated IPv6 CIDR block', 'Amazon-provided IPv6 CIDR block', and 'IPv6 CIDR owned by me' are unselected.
- Tenancy:** A dropdown menu with 'Default' selected.

Create one-subnet

vpc=custom vpc(vpc-1)

Subnet name=public-subnet-1a

Availability Zone=1a

IPv4 VPC CIDR block=10.0.0.0/16

IPv4 subnet CIDR block=10.0.0.0/24(256 ip) are given to this subnet for association
create...

[EC2](#) > [Subnets](#) > Create subnet

VPC ID
Create subnets in this VPC.

vpc-058a97e6e09353468 (vpc-1) ▼

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

public-subnet-1a

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-s1-az1 (ap-south-1a) ▼

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16 ▼

IPv4 subnet CIDR block

10.0.0.0/24 256 IPs

< > ^ v

Create ec2 instance
Name and tags=linuxp
Application and OS Images=ubutnu
Instance type=t3.micro
>>.. Create key-pair for instance

Name and tags [Info](#)

Name
linuxp [Add additional tags](#)

Application and OS images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents [Quick Start](#)

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), 55D Volume Type Free tier eligible

ami-02b8269d5e85954ef (64 bit x86) / ami-02730d8f73a05d22c (64 bit x86_64) / ami-02b8269d5e85954ef (ARMv8, 64 bit ARM) / ami-02b8269d5e85954ef (ARMv8, 64 bit ARM)

Description

Ubuntu Server 24.04 LTS (HVM), 55D Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture AMI ID **Publish Date** **Username**

64-bit (x86) ami-02b8269d5e85954ef 2025-10-22 ubuntu Verified provider

Instance type [Info](#) [Get advice](#)

Instance type

t3.micro Free tier eligible

Family: t3 2 vCPU 1 GB Memory Current generation free On-Demand Linux base pricing: 0.0112 USD per Hour

On-Demand SUSE base pricing: 0.0112 USD per Hour On-Demand Windows base pricing: 0.0244 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour

☒ All generations [Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

public-1a-key [Create new key pair](#)

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64 noble image
ami-02b8269d5e85954ef

Virtual server type (Instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Create key pair [X](#)

Key pair name

Key pairs allow you to connect to your instance securely.

public-1a-key

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

[Cancel](#) [Create key pair](#)

Network settings=Edit

VPC - required= 1a

Subnet=public-1a

Auto-assign public IP=enable

Inbound Security Group Rules

>>...Type=http

Source type= anywhere

Network settings

VPC - required | [Info](#)

vpc-058a97efe09353468 (vpc-1)
10.0.0.0/16

Subnet | [Info](#)

subnet-0c14001e02c625846
VPC: vpc-058a97efe09353468
Zone type: Availability Zone

Owner: 601920987231
Availability Zone: ap-south-1a (aps1-az1)
IP addresses available: 251
CIDR: 10.0.0.0/24

Create new subnet

Auto-assign public IP | [Info](#)

Enable

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

launch-wizard-3

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _; / [! @ # % & *] { } | ~

Description - required | [Info](#)

launch-wizard-3 created 2026-01-18T16:29:34.695Z

Inbound Security Group Rules

Security group rule 1 (TCP; 22, 0.0.0.0/0)

Remove

Type | [Info](#)

ssh

Protocol | [Info](#)

TCP

Port range | [Info](#)

22

Source type | [Info](#)

Anywhere

Source | [Info](#)

Description - optional | [Info](#)

e.g. SSH for admin desktop

Security group rule 2 (TCP; 80, 0.0.0.0/0)

Remove

Type | [Info](#)

HTTP

Protocol | [Info](#)

TCP

Port range | [Info](#)

80

Source type | [Info](#)

Anywhere

Source | [Info](#)

Description - optional | [Info](#)

e.g. SSH for admin desktop

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Add security group rule

Summary

Number of instances | [Info](#)

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-02b8269d5e68594ef

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

launch instance.. (at this moment our ec2 instance will not work because rt,internet gateway are created)

Create internet gateway

Name=IGW-1

create...

Create internet gateway

[Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

IGW-1

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Remove

Add new tag

You can add 49 more tags.

Cancel

Create internet gateway

Attach to vpc...

Attach to VPC (igw-0199021f514aaf155) [info](#)

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

► **AWS Command Line Interface command**

[Cancel](#) [Attach internet gateway](#)

Create route table name=rt-1 vpc=vpc-1a

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

Edit subnet associations and save it

Edit subnet associations
Change which subnets are associated with this route table.

Available subnets (1/1)

☒	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	public-subnet-1a	subnet-0c14001e02c623846	10.0.0.0/24	-	Main (rtb-0f47b4e910cb9eda0)

Selected subnets

[Cancel](#) [Save associations](#)

Routes
route=anywhere
Source type=internet gateway
Name=IGW-1
save...

VPCC > Route tables > rtb-0a6e4ea26fd928435 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin	
10.0.0.0/16	local	Active	No	CreateRouteTable	
0.0.0.0/0	Internet Gateway	-	No	CreateRoute	Remove

[Add route](#) [Cancel](#) [Preview](#) [Save changes](#)

Try and run your ec2 instance it will run because your ec2=subnet=igw=rt are all connected together

To run a command as administrator (user "root"), use "sudo <command>". See "man sudo_root" for details.

```
ubuntu@ip-10-0-0-44:~$ ls
ubuntu@ip-10-0-0-44:~$ which bash
/usr/bin/bash
ubuntu@ip-10-0-0-44:~$
```

Create 2 vpc

Vpc-2

cidr(classless inter domain name)=192.168.0.0/16 (65,536) ip address

Tendency=default

Create...

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

vpc-2

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
192.168.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)
Default

VPC encryption control (\$) [Info](#)
Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. [Additional charges apply](#)

☒ None ☐ Monitor mode ☐ Enforce mode

Create subnet for 2 vpc

vpc=custom vpc(vpc-1)

Subnet name=prod-1b

Availability Zone=1b

IPv4 VPC CIDR block=192.168.0.0/16

IPv4 subnet CIDR block=192.168.0.0/24(256 ip) are given to this subnet for association create...

[VPC](#) > [Subnets](#) > Create subnet

VPC ID
Create subnets in this VPC.

vpc-0e06a6d5ef5a364d9 (vpc-2) ▼

Associated VPC CIDRs

IPv4 CIDRs
192.168.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

prod-1b

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1b ▼

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/16 ▼

IPv4 subnet CIDR block

192.168.0.0/24 256 IPs

< > ^ v

Create internet gateway name=prod-IGW

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

prod-IGW

Attach after creating a vpc

Attach to VPC (igw-05ee87fb8a0928f12) [Info](#)

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.

Q vpc-0e06a6d5ef5a364d9

► [AWS Command Line Interface command](#)

[Cancel](#)

[Attach internet gateway](#)

Create route table for and associate it with second vpc

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

prod-1

VPC

The VPC to use for this route table.

vpc-0e06a6d5ef5a364d9 (vpc-2)

Available subnets (1/1)

Q Filter subnet associations

☒	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
☒	prod-1b	subnet-07208a20f76a21095	192.168.0.0/24	-

Selected subnets

subnet-07208a20f76a21095 / prod-1b

[Cancel](#)

[Save associations](#)

Destination	Target	Status	Propagated	Route Origin
192.168.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0	Q local			
	Internet Gateway	-	No	CreateRoute
	Q igw-05ee87fb8a0928f12			
Add route				

[Cancel](#)

[Preview](#)

[Save changes](#)

Create ec2 instance

▼ Network settings | Info

VPC - required | Info

vpc-0e06a6d5ef5a364d9 (vpc-2)
192.168.0.0/16

Subnet | Info

subnet-07208a20f76a21095
VPC: vpc-0e06a6d5ef5a364d9
Zone type: Availability Zone
Owner: 601920987231
Availability Zone: ap-south-1b (aps1-a-z3)
IP addresses available: 251
CIDR: 192.168.0.0/24

Auto-assign public IP | Info

Enable

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group | Select existing security group

Security group name - required

launch-wizard-4

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./[!@#%&'()*+,-=:;{}|~]*

Description - required | Info

launch-wizard-4 created 2026-01-18T17:24:47.165Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP; 22, 0.0.0.0/0)

Type | Info

ssh

Protocol | Info

TCP

Port range | Info

22

Source type | Info

Anywhere

Source | Info

Q Add CIDR, prefix list or security group

0.0.0.0/0 X

Description - optional | Info

e.g. SSH for admin desktop

▼ Security group rule 2 (TCP; 80, Multiple sources)

Type | Info

HTTP

Protocol | Info

TCP

Port range | Info

80

Source type | Info

Custom

Source | Info

Q Add CIDR, prefix list or security group

0.0.0.0/0 X X

Description - optional | Info

e.g. SSH for admin desktop

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Add security group rule

▼ Summary

Number of instances | Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more
ami-02b82694d5e85954ef

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Create peering between both vpc
name=prod-test-peering
vpc=select vpc in which peering is created
account=my account
region=same region
vpc=second vpc to accepted the peering request

VPC > Peering connections > Create peering connection

Peering connection settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

prodt-test-peering

Select a local VPC to peer with

VPC ID (Requester)
vpc-058a97e6e09353468 (vpc-1)

VPC CIDRs for vpc-058a97e6e09353468 (vpc-1)

CIDR	Status	Status reason
10.0.0.0/16	Associated	-

Select another VPC to peer with

Account
☒ My account
☐ Another account

Region
☒ This Region (ap-south-1)
☐ Another Region

VPC ID (Acceptor)
vpc-0e06a6d5ef5a364d9 (vpc-2)

VPC CIDRs for vpc-0e06a6d5ef5a364d9 (vpc-2)

CIDR	Status	Status reason
192.168.0.0/16	Associated	-

Accept vpc peering request on second vpc

peering

Tags

Accept VPC peering connection request [Info](#)

Are you sure you want to accept this VPC peering connection request? (pcx-0e69366d6a8a07316 / prodt-test-peering)

Requester VPC vpc-058a97e6e09353468 / vpc-1	Acceptor VPC vpc-0e06a6d5ef5a364d9 / vpc-2	Requester CIDRs 10.0.0.0/16
Acceptor CIDRs -	Requester Region Mumbai (ap-south-1)	Acceptor Region Mumbai (ap-south-1)
Requester owner ID 601920987231(This account)	Acceptor owner ID 601920987231(This account)	

Cancel
Accept request

Edit route table

Cidr prod ip=192.168.0.0/16 in 1 vpc (test) to second and vise-verse and both come under peering connection. And save...

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute
192.168.0.0/16	igw-0199021f514aaf155	-	No	CreateRoute
	Peering Connection	-	No	CreateRoute
	pcx-0e69366d6a8a07316	-	No	CreateRoute

[Add route](#) [Cancel](#) [Preview](#) [Save changes](#)

Both cidr to peerin connetction

Edit routes

Destination	Target	Status	Propagated	Route Origin
192.168.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute
10.0.0.0/16	igw-05ee87fb8a0928f12	-	No	CreateRoute
	Peering Connection	-	No	CreateRoute
	pcx-0e69366d6a8a07316	-	No	CreateRoute

[Add route](#) [Cancel](#) [Preview](#) [Save changes](#)

Editing security group to ping the machings

Test ec2>security>inbound

Add role= ICMP IPV4

Custom = test mai prod ka cidr range(192.168.0.0/16)

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sgr-01b31270be81ad5c8	SSH	TCP	22	Custom	0.0.0.0/0
sgr-047c0f318d2d43610	HTTP	TCP	80	Custom	0.0.0.0/0
-	All ICMP - IPv4	ICMP	All	Custom	192.168.0.0/16

[Add rule](#) [Cancel](#) [Preview changes](#) [Save rules](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Prod-test

Test ec1>security>inbound

Add role= ICMP IPV4

Custom = test mai prod ka cidr range(10.0.0.0/16)

EC2 > Security Groups > sg-021c901fc4405e7d7 - launch-wizard-5 > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sg-07cbdddec6bd230b4a	HTTP	TCP	80	Custom	Q	Delete
sg-0258e5bede4e3d60d	SSH	TCP	22	Custom	Q 0.0.0.0/X	Delete
-	All ICMP - IPv4	ICMP	All	Custom	Q 0.0.0.0/X Q 10.0.0.0/16 Q 10.0.0.0/16/X	Delete

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Cancel](#) [Preview changes](#) [Save rules](#)

Connect to ec2 machine and try to ping ec1
 Select EC2 machine and connect it
 Ping ec2 test machine to ping prod ip
 Ping 10.0.0.44.

```
Last login: Sun Jan 18 18:11:28 2026 from 13.233.177.3
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-192-168-0-49:~$ ls
ubuntu@ip-192-168-0-49:~$ ping 10.0.0.44
PING 10.0.0.44 (10.0.0.44) 56(84) bytes of data:
64 bytes from 10.0.0.44: icmp_seq=1 ttl=64 time=0.497 ms
64 bytes from 10.0.0.44: icmp_seq=2 ttl=64 time=0.552 ms
64 bytes from 10.0.0.44: icmp_seq=3 ttl=64 time=0.529 ms
64 bytes from 10.0.0.44: icmp_seq=4 ttl=64 time=0.543 ms
64 bytes from 10.0.0.44: icmp_seq=5 ttl=64 time=0.514 ms
64 bytes from 10.0.0.44: icmp_seq=6 ttl=64 time=0.534 ms
64 bytes from 10.0.0.44: icmp_seq=7 ttl=64 time=0.497 ms
64 bytes from 10.0.0.44: icmp_seq=8 ttl=64 time=0.522 ms
64 bytes from 10.0.0.44: icmp_seq=9 ttl=64 time=0.513 ms
64 bytes from 10.0.0.44: icmp_seq=10 ttl=64 time=0.508 ms
64 bytes from 10.0.0.44: icmp_seq=11 ttl=64 time=0.517 ms
64 bytes from 10.0.0.44: icmp_seq=12 ttl=64 time=0.544 ms
64 bytes from 10.0.0.44: icmp_seq=13 ttl=64 time=0.511 ms
64 bytes from 10.0.0.44: icmp_seq=14 ttl=64 time=0.504 ms
64 bytes from 10.0.0.44: icmp_seq=15 ttl=64 time=0.525 ms
64 bytes from 10.0.0.44: icmp_seq=16 ttl=64 time=0.516 ms
64 bytes from 10.0.0.44: icmp_seq=17 ttl=64 time=0.491 ms
64 bytes from 10.0.0.44: icmp_seq=18 ttl=64 time=0.536 ms
64 bytes from 10.0.0.44: icmp_seq=19 ttl=64 time=0.541 ms
```

